

Variance Swap Exposure

Extension to the Project *Hedging Analysis with Greeks*

Project Summary

This project extends the project *Hedging Analysis with Greeks* framework from delta-hedged vanilla call options to instruments whose payoff is directly tied to realized variance. This framework mirrors how variance swaps and VIX are priced and hedged on volatility desks.

It develops the connections from three perspectives:

- 1) the continuous-time log-contract replication identity;
- 2) the trading interpretation through a dynamically delta-hedged short log contract;
- 3) the static pricing of the fair variance strike using the OTM option strip

The numerical experiments include discrete realized-variance estimators, normalized dollar-gamma exposures, and a VIX-style discretization of the continuous option-strip formula.

The results show:

- 1) Across 5000 GBM paths with $\sigma = 0.2$, the mean squared-log-return variance is 0.03999688, compared with 0.03999578 from the discrete portfolio replication estimator and the theoretical value 0.04.
- 2) The mean replication error is $-1.10 \cdot 10^{-6}$, with RMSE $4.02 \cdot 10^{-5}$.
- 3) After equalizing initial variance exposure, the vanilla call and the short log contract begin with the same dollar-gamma weight. The call's subsequent exposure changes with moneyness and remaining maturity, while the short log contract retains a constant weight.
- 4) Carr-Madan weights mechanically assigns greater weight to lower strikes. Actual strike contributions depend on the product of option strip and weights. Under a real negative skew typically observed in equity-index options, the greater contribution of downside puts can be quantified.

Table of Contents

1.	Two Views of the Realized-variance Mechanism	1
1.1	Derivation of Variance Swap Replication Identity	1
1.2	View #1: Mathematical Identity Check	2
1.3	View #2: Portfolio Construction	2
1.4	Realized Log-return Variance	3
1.5	Simulations	3
2.	Vanilla Call Option Gamma P&L vs Log Contract and Variance Swap P&L	7
2.1	Per-step P&L Approximations for Vanilla Call Option and Log Contract	7
2.2	Simulation Results	8
3.	Static Option-strip Approximation of the Fair Variance Strike	11
3.1	Derivation of Variance Strike Fair Value	11
3.2	Mathematical Derivation of Discrete Form for Numerical Analysis	13
3.3	Simulations	14
4.	Future Extensions	15

In the main *Hedging Analysis with Greeks* project, P&L for a delta-hedged vanilla call option depends on dollar gamma exposure. Here this project extends that framework to instruments whose payoff is directly tied to realized variance, namely log contracts and variance swaps. This allows us to reinterpret the variance-spread P&L of a delta-hedged option as a state-dependent weighting of an idealized variance swap.

1. Two Views of the Realized-variance Mechanism

This project implements two approaches checking the realized log-return variance: one is mathematical identity verification, and another is to construct a portfolio with trading interpretations. Also, it uses the usual realized log-return variance formula as the benchmark.

1.1 Derivation of Variance Swap Replication Identity

The underlying is driven by geometric Brownian motion: $dS_t = \mu_t S_t dt + \sigma_t S_t dX_t$, where dX_t is the Brownian increment.

according to Itô lemma,

$$dV(S_t, t) = \frac{\partial V}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S_t^2} dS_t^2$$

Since, $dS_t^2 = \sigma_t^2 S_t^2 dX_t^2 = \sigma_t^2 S_t^2 dt^1$, then:

$$dV(S_t, t) = \frac{\partial V}{\partial S_t} dS_t + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} dt$$

Here, $V(S_t, t) = \log S_t$, $\log(\cdot)$ function is nonlinear, and concave.

$$\frac{\partial V}{\partial S_t} = \frac{1}{S_t}, \quad \frac{\partial^2 V}{\partial S_t^2} = -\frac{1}{S_t^2}$$

$$\begin{aligned} d\log S_t &= \frac{1}{S_t} dS_t + \frac{1}{2} \sigma_t^2 S_t^2 \left(-\frac{1}{S_t^2} \right) dt \\ &= \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 dt \end{aligned}$$

$-\frac{1}{2} \sigma_t^2 dt$ is the Itô correction, curvature/convexity adjustment. Therefore:

$$\frac{dS_t}{S_t} - d\log S_t = \frac{1}{2} \sigma_t^2 dt$$

¹ Using the quadratic variation of Brownian motion, and higher-order terms, squared and cross terms, vanish.

The integral form is:

$$\int_0^T \frac{dS_t}{S_t} - \log \frac{S_T}{S_0} = \frac{1}{2} \int_0^T \sigma_t^2 dt$$

This formula is the variance swap replication identity. The identity from Itô lemma is exact in continuous time and replicates $\int_0^T \sigma_t^2 dt$.

$\int_0^T \sigma_t^2 dt$ is the integrated realized variance, log-price quadratic variation, $\langle \log S_t \rangle_T$

1.2 View #1: Mathematical Identity Check

It is the numerical convergence check, validating the discrete approximation to the continuous identity.

The discrete form of variance swap replication identity is:

$$RV_{rep} = 2 \sum_i \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} - 2 \log \frac{S_T}{S_0}$$

The discrete increment is:

$$RV_{rep,i} = 2 \left[\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} - \log \frac{S_i}{S_{i-1}} \right]$$

1.3 View #2: Portfolio Construction

The LHS of variance swap replication identity is the portfolio of traded instruments: short log contract and dynamic constant-dollar stock hedging.

$$V(S_t, t) = \log S_t, \text{ or } V(S_t, t) = \log \frac{S_t}{S_0}$$

$$d \log S_t = \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 dt \text{ (Itô lemma)}$$

Under the assumption of zero interest rates and dividends, the delta-hedged portfolio simplifies to

$$\pi = -\log S_t + \Delta_t S_t, \text{ where } \Delta_t = \frac{1}{S_t} \text{ (omits the cash account),}$$

yielding P&L for per time step:

$$d\pi = -d \log S_t + \frac{1}{S_t} dS_t$$

$$\begin{aligned}
&= -\frac{dS_t}{S_t} + \frac{1}{2}\sigma_t^2 dt + \frac{1}{S_t}dS_t \\
&= \frac{1}{2}\sigma_t^2 dt
\end{aligned}$$

Recall $RV_{rep,i} = 2 \left[\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} - \log \frac{S_i}{S_{i-1}} \right]$ in 1.2, from portfolio construction perspective, It can be interpreted as the discrete portfolio gain, the exact gain of the discretely rebalanced portfolio over the interval $[t_{i-1}, t_i]$, ignoring financing and the fixed variance leg. It is the discretely rebalanced log-contract estimator, the finite-grid portfolio expression.

1.4 Realized Log-return Variance

The benchmark is the discrete form of realized log-return variance. The formula is:

$$RV = \sum_i \left(\log \frac{S_{t_i}}{S_{t_{i-1}}} \right)^2$$

The discrete increment is:

$$RV_{log,i} = \left(\log \frac{S_{t_i}}{S_{t_{i-1}}} \right)^2$$

It is the standard discretely monitored realized variance.

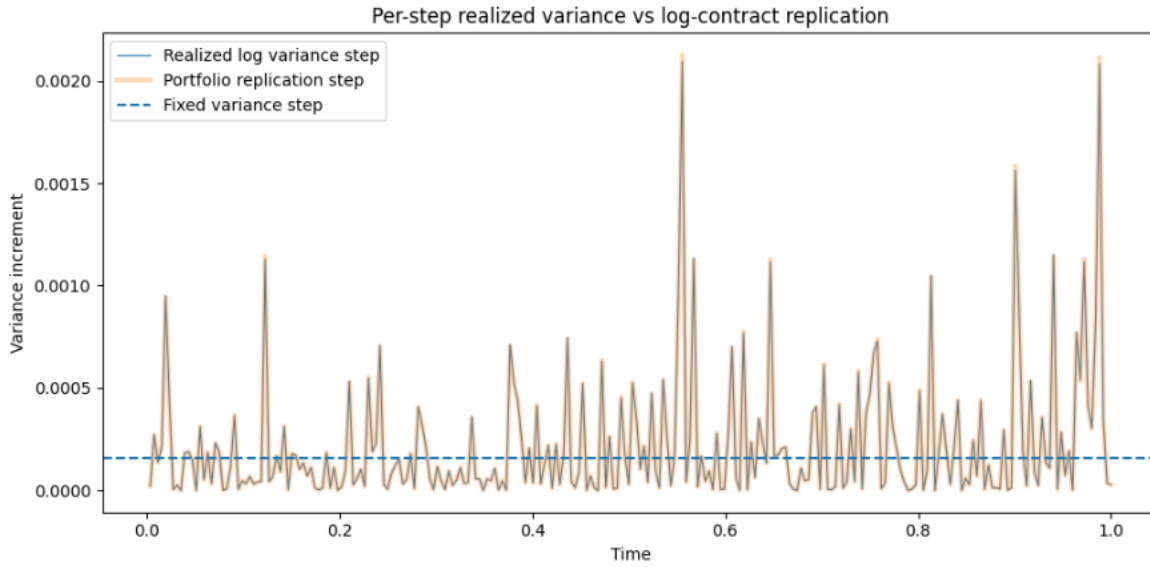
1.5 Simulations

Mean log realized variance (benchmark)	0.03999688
Mean replicated variance (realized variance discrete form)	0.03999578
Theoretical integrated variance	0.04000000
Mean replication error	-1.10E-06
RMSE replication error	4.02E-05
Mean delta-hedged log P&L (portfolio construction)	0.03999578
Mean difference	-1.10E-06
RMSE error	4.02E-05

Scenario: $S_0 = 100$, $\mu = 0.0$, $\sigma = 0.2$, $T = 1.0$, steps = 252, paths = 5000

The results show that the mean replication error is on the order of 10^{-6} , and the RMSE is $\sim 4e-5^2$. It is small relative to the variance level and decreases with finite monitoring. Across 5000 simulated paths, the discretized replication strategy approximates the continuous-time benchmark at 252 monitoring dates.

For the single path simulation, the results are shown below:



Scenario: $S_0 = 100$, $\mu=0.0$, $\sigma = 0.25$, $T = 1.0$, steps = 252

Notes:

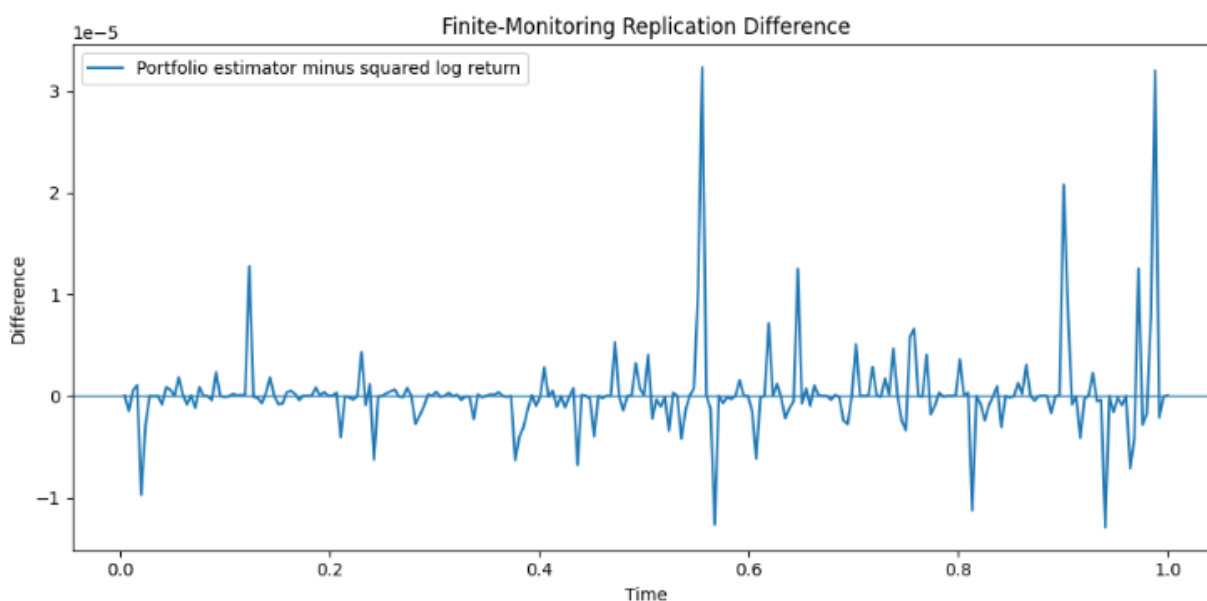
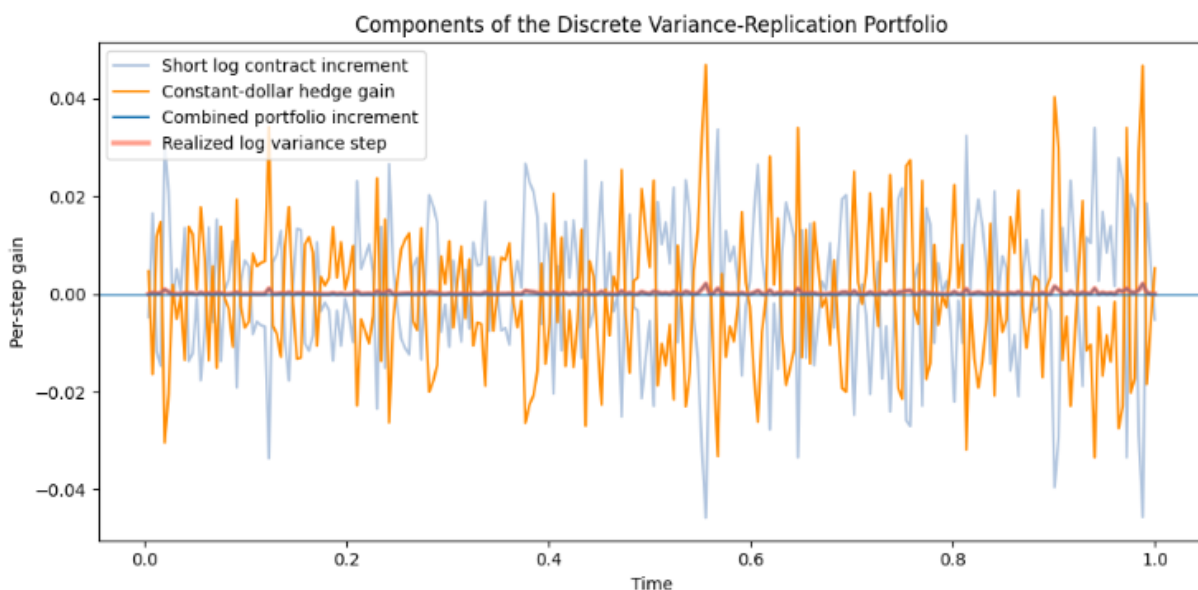
Realized log variance step: variance benchmark, squared log return, $RV_{log,i}$

Portfolio replication step: discrete portfolio increment, short log-contract + constant dollar hedge, $RV_{rep,i}$

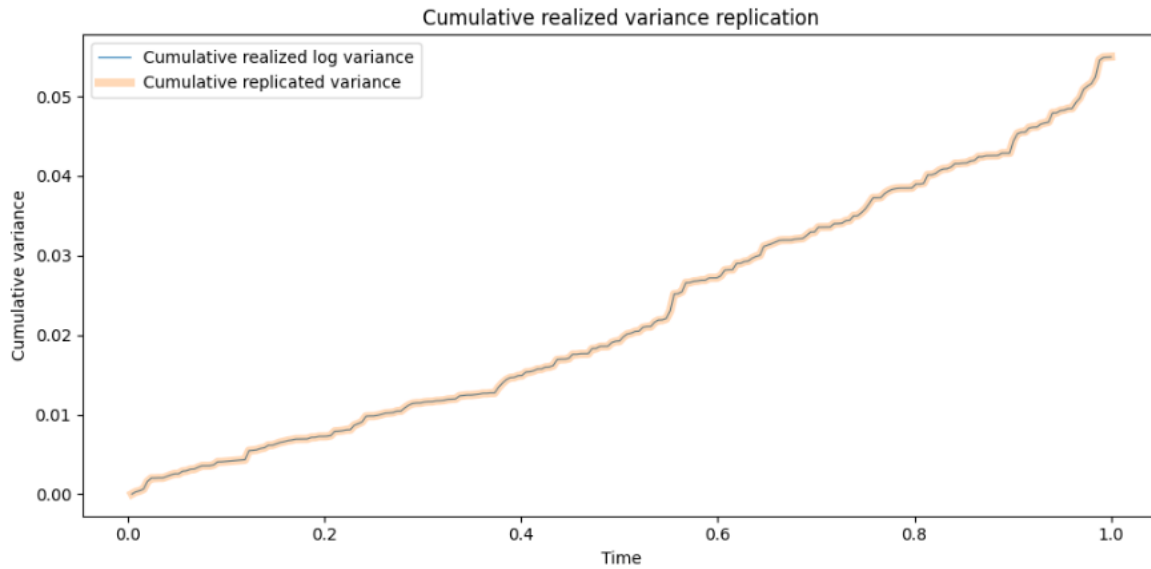
The single-path plots show that the discrete portfolio estimator tracks squared log returns closely on a path-by-path basis. The two estimators are not exactly identical at finite monitoring frequency. They differ by higher-order terms at finite dt , while their difference converges to zero as the grid becomes finer.

The single-path plots illustrate that the discrete replication converges pathwise under grid refinement. It is not just in expectation.

² The underlying path simulation is built by using exact lognormal increments, rather than Euler-Maruyama or Milstein schemes. Therefore, the residual here $\sim 4e-5$ is coming purely from the sum-of-squared-log-returns quadratic-variation estimator, not from any error in simulating the price path itself.

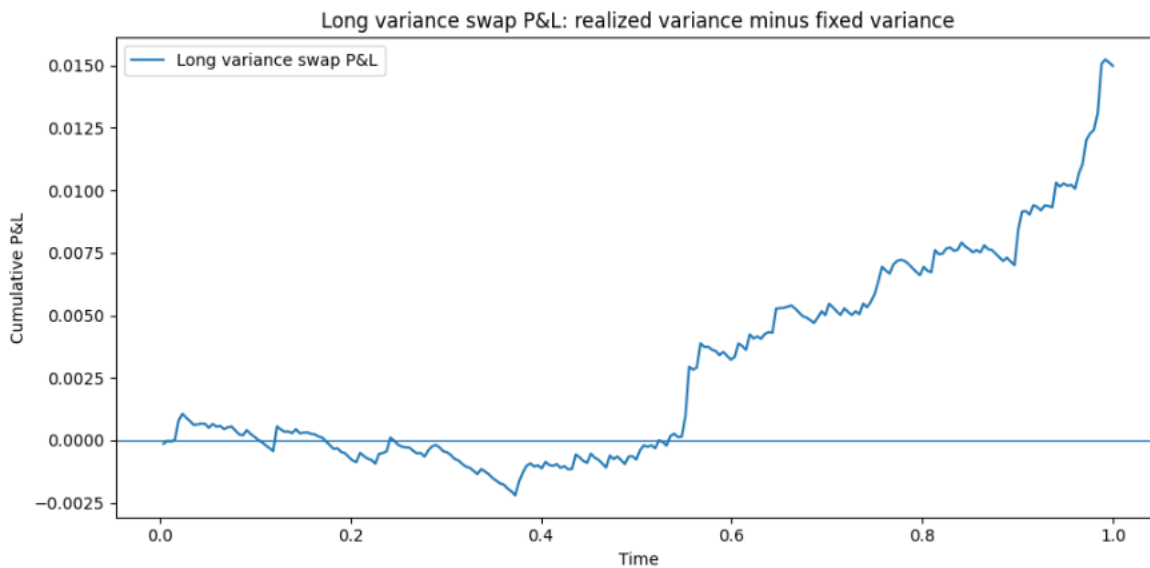


The plots illustrate the short log contract increment and constant-dollar stock-hedge gain are approximately mirror-image first-order exposures. When these two components are combined, the first-order return terms cancel almost exactly, leaving a small second-order convexity gain that approximates the squared log return. The finite-monitoring difference arises from higher-order return terms. It is clear that large positive or negative returns generate the visible spikes in the finite-monitoring difference.



Scenario: $S_0 = 100$, $\mu=0.0$, $\sigma = 0.25$, $T = 1.0$, steps = 252

The plot above shows that two cumulative series match from start to finish, enforcing the replication formula holds at the cumulative level.



Scenario: $S_0 = 100$, $\mu=0.0$, $\sigma = 0.25$, implied $\sigma = 0.2$, $T = 1.0$, steps = 252, paths = 5000

The plot above shows the synthetic variance swap P&L, by subtracting a fixed variance strike from the cumulative replicated variance.

Final realized log variance	0.054926
Final replicated variance	0.054984

Replication error	5.80E-05
Final variance swap P&L	0.014984

The single path simulation shows the error 5.8e-5, compared to ~4.0e-5 in the 5000-path run.

2. Vanilla Call Option Gamma P&L vs Log Contract and Variance Swap P&L

2.1 Per-step P&L Approximations for Vanilla Call Option and Log Contract

For delta-hedged vanilla call option, the P&L over time step dt is:

$$d\pi_{vanilla\ call} = \frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt$$

σ_a : actual/realized volatility

σ_i : implied volatility

where $\frac{1}{2} S^2 \Gamma_i$ is gamma weight. The weight changes with spot, moneyness, and time-to-maturity (refer to project *Hedging Analysis with Greeks*).

For the log contract, $V(S_t, t) = \log S_t$:

$$\frac{d^2 V}{ds_t^2} = -\frac{1}{s_t^2}, \quad \frac{1}{2} S^2 \Gamma_i = \frac{1}{2} S^2 * \left(-\frac{1}{s_t^2}\right) = -\frac{1}{2}$$

So, for short position of the log contract, the P&L over time step dt is:

$$d\pi_{short\ log} = \frac{1}{2} (\sigma_a^2 - \sigma_i^2) dt$$

The short position of the log contract makes the position convex.

Unlike the gamma-weighted variance exposure for a vanilla call option, the short log contract shows pure variance-spread exposure.

It is the theoretical drift approximation. If realized variance is used in the simulation,

namely, $\left(\log \frac{S_{t_i}}{S_{t_{i-1}}}\right)^2$ in the formula (refer to the discrete form of realized log-return

variance), the P&L over time step dt is:

$d\pi_{short\ log} = \frac{1}{2}((d\log S_t)^2 - \sigma_i^2 dt)$ (comparing to $\frac{1}{2}\Gamma S^2 \left[\left(\frac{dS}{S}\right)^2 - \sigma_i^2 dt \right]$ for vanilla call option)

2.2 Simulation Results

For a fair comparison, it is essential to normalize the vanilla call notional so that its initial variance exposure equals the log-contract exposure.

Weight of the vanilla call: $w_{call} = \frac{1}{2}\Gamma S^2$

Weight of short log contract: $w_{short\ log} = \frac{1}{2}$, for one-unit log contract

Normalized weight of the vanilla call should be $\frac{1}{2}$, $\widetilde{w_{call,t=0}} = \frac{1}{2}$, which means they begin with the same variance exposure at inception.

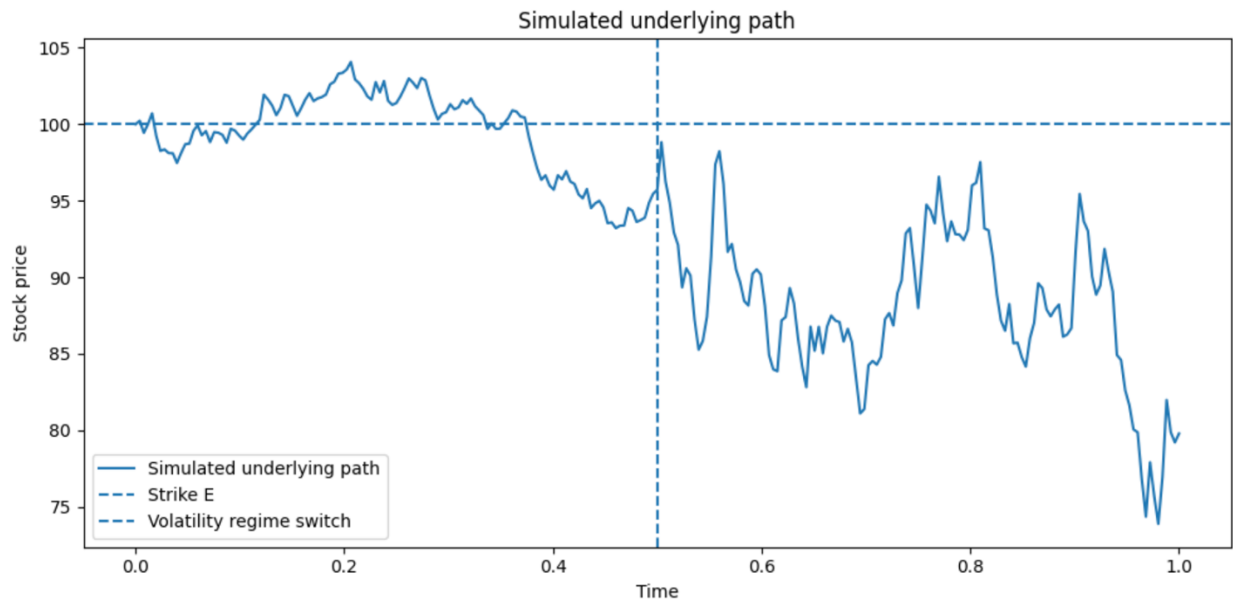
So, $N_{call}w_{call} = \widetilde{w_{call,t=0}}$, the scaled call exposure.

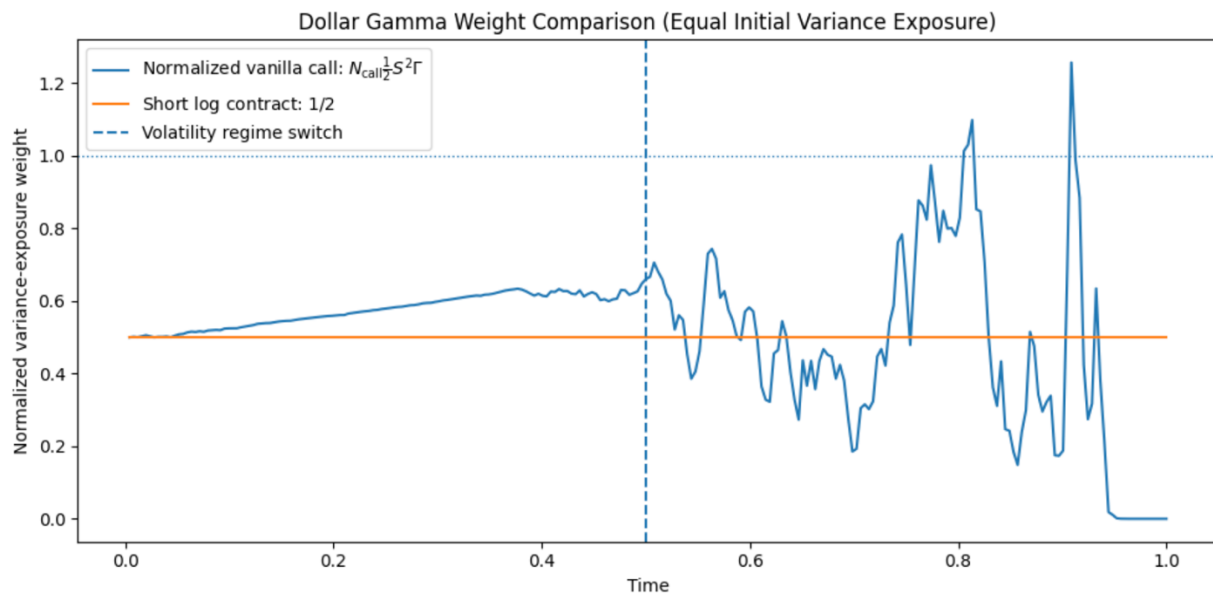
N_{call} : number of vanilla call option held

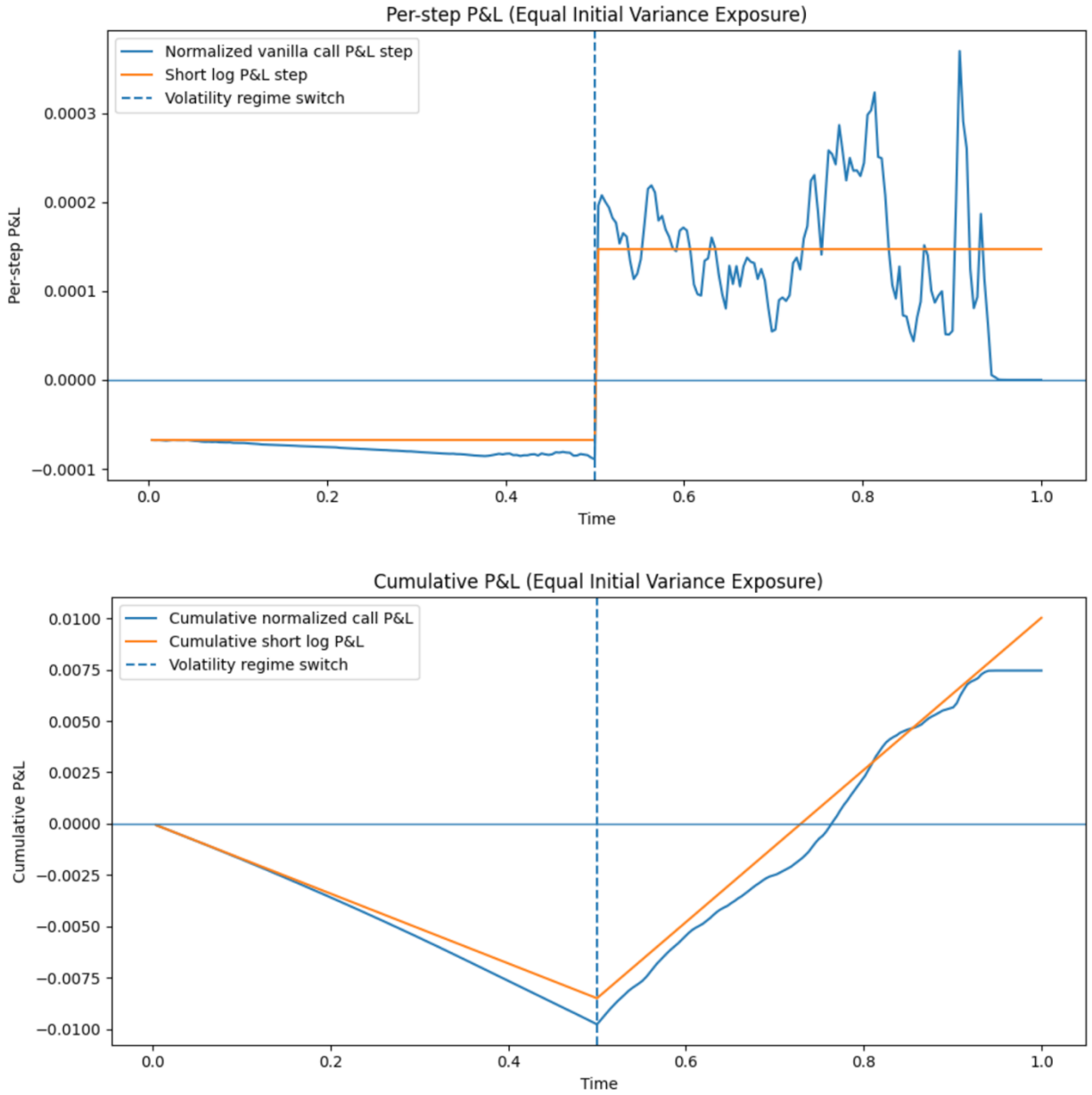
$N_{call} = \frac{1}{2w_{call,t=0}}$, at time $t = 0$

$$\widetilde{w_{call,t}} = \frac{1}{2w_{call,t=0}} * w_{call}$$

The following plots show the comparison of dollar gamma weight, P&L over time step and accumulation based on vanilla call option and short log contract.







Scenario: $S_0 = 100$, $\mu = 0.0$, $\sigma\text{-low} = 0.12$, $\sigma\text{-high} = 0.35$, $\text{implied-sigma} = 0.22$, $E = 100$, $T = 1.0$, $\text{steps} = 252$

It is a clear and direct visual display that the log-contract's dollar-gamma weight is constant at $\frac{1}{2}$ across the whole path, while the vanilla call's dollar-gamma weight is concentrated around the strike. As expiry approaches, it collapses when spot remains away from the strike but can spike sharply when spot is close to the strike. Its delta-

hedged P&L is path-dependent. In contrast, the log contract's constant instantaneous dollar-gamma exposure indicates it is a pure bet on realized variance under assumption of continuous-diffusion and continuous-rebalancing. The short log contract removes the state-dependent dollar-gamma weighting of vanilla option. Its exposure is independent of moneyness, although its realized P&L still depends on the accumulated realized variance path.

This is the theoretical reason variance swaps can be replicated model-free, while volatility exposure via vanilla options is inherently local and model-dependent.

3. Static Option-strip Approximation of the Fair Variance Strike

3.1 Derivation of Variance Strike Fair Value

To replicate the variance payoff statically, it is essential to determine the risk-neutral probability density of the asset price at maturity.

According to Breeden-Litzenberger result, PDF is

$$p(S_T = E) = \frac{d^2 \widetilde{V}_{call}(E, T)}{dE^2} = \frac{d^2 \widetilde{V}_{put}(E, T)}{dE^2}$$

$\widetilde{V}_{call}(S_t, E, t, T)$: undiscounted call price

$\widetilde{V}_{put}(S_t, E, t, T)$: undiscounted put price

The second strike derivative reveals the risk-neutral terminal density.

The payoff formula is

$$g(S_T) = g(F) + g'(F)(S_T - F) + \int_0^F g''(E) (E - S_T)^+ dE + \int_F^\infty g''(E) (S_T - E)^+ dE$$

The decomposition of a nonlinear payoff is: 1) cash; 2) a forward position; 3) a continuum of put strip below the forward; 4) a continuum of call strip above the forward
 $g''(E)$ is the weight, curvature of the payoff at strike E

According to continuous expectation formula,

$$\mathbb{E}[g(S_T)|S_t] = \int_0^F \frac{d^2 \widetilde{V}_{put}(E, T)}{dE^2} g(E) dE + \int_F^\infty \frac{d^2 \widetilde{V}_{call}(E, T)}{dE^2} g(E) dE$$

Integrating by parts twice and using the put-call parity relation:

$$\mathbb{E}[g(S_T)|S_t] = g(F) + \int_0^F \widetilde{V}_{put}(E) g''(E) dE + \int_F^\infty \widetilde{V}_{call}(E) g''(E) dE$$

It is the static-replication formula, requires no rebalancing, and is model-independent.

For a log contract, payoff $g(S_T) = \log\left(\frac{S_T}{F}\right)$

$g(F) = 0$, $g''(E) = -\frac{1}{E^2}$. So:

$$\mathbb{E}[g(S_T)|S_t] = \mathbb{E}\left[\log\left(\frac{S_T}{F}\right)\right] = -\int_0^F \widetilde{V}_{put}(E) \frac{1}{E^2} dE - \int_F^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE$$

The negative sign results from the concavity of log payoff, $g''(E) < 0$. The short position of the log contract is convex.

Again, refer to the variance swap replication identity:

$$\int_0^T \frac{dS_t}{S_t} - \log \frac{S_T}{S_0} = \frac{1}{2} \int_0^T \sigma_t^2 dt$$

Taking expectation under the risk-neutral measure on both sides:

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \frac{dS_t}{S_t} \right] - \mathbb{E}^{\mathbb{Q}} \left[\log \frac{S_T}{S_0} \right] = \frac{1}{2} \mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right]$$

Under the assumption that interest rate and dividends are zero, and under the risk-neutral measure:

$$\int_0^T \frac{dS_t}{S_t} = \int_0^T \sigma_t dX_t^{\mathbb{Q}}, \text{ It is a martingale.}$$

$$\text{So, } \mathbb{E}^{\mathbb{Q}} \left[\int_0^T \frac{dS_t}{S_t} \right] = 0$$

$$\text{Getting } \mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right] = -2 \mathbb{E}^{\mathbb{Q}} \left[\log \frac{S_T}{S_0} \right]$$

Applying the static option strip replication formula (the log contract provides the connection),

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right] = 2 \left[\int_0^F \widetilde{V}_{put}(E) \frac{1}{E^2} dE + \int_F^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE \right]$$

The variance swap payoff is $N_{var}(RV_T - E_{var})$

N_{var} : variance notional

RV_T : floating leg, ex-post realized variance

E_{var} : fixed leg, pre-agreed at inception, variance level, reference level, variance strike

At inception, the fair price of a variance swap is zero, so the variance strike equals the risk-neutral expectation of realized variance.

$$E_{var} = \mathbb{E}^{\mathbb{Q}}[RV_T] = \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{T}\int_0^T \sigma_t^2 dt\right]$$

$$\text{Therefore, } E_{var} = \frac{2}{T}\left[\int_0^F \widetilde{V}_{put}(E) \frac{1}{E^2} dE + \int_F^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE\right]$$

3.2 Mathematical Derivation of Discrete Form for Numerical Analysis

The idea of VIX construction is a discretized, practical application of the continuous log-strip formula used for variance -swap pricing.

According to the variance strike formula above:

$$VIX^2 = \frac{2}{T}\left[\int_0^F \widetilde{V}_{put}(E) \frac{1}{E^2} dE + \int_F^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE\right]$$

$$VIX^2 \frac{T}{2} = \int_0^F \widetilde{V}_{put}(E) \frac{1}{E^2} dE + \int_F^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE$$

VIX methodology defines E_0 as the highest listed strike below F :

$$VIX^2 \frac{T}{2} = \int_0^{E_0} \widetilde{V}_{put}(E) \frac{1}{E^2} dE + \int_{E_0}^\infty \widetilde{V}_{call}(E) \frac{1}{E^2} dE + \int_{E_0}^F \left(\widetilde{V}_{put}(E) - \widetilde{V}_{call}(E)\right) \frac{1}{E^2} dE$$

The first two items are the full OTM-option strip: $\int_0^\infty Q(E) \frac{1}{E^2} dE$

Then, using the put-call parity, and approximation $E^2 \approx E_0^2$

$$VIX^2 \frac{T}{2} \approx \int_0^\infty Q(E) \frac{1}{E^2} dE - \frac{(F - E_0)^2}{2E_0^2}$$

Writing the integral as the discrete form: $\sum_i \frac{\Delta E_i}{E_i^2} Q_i(E_i)$

Finally,

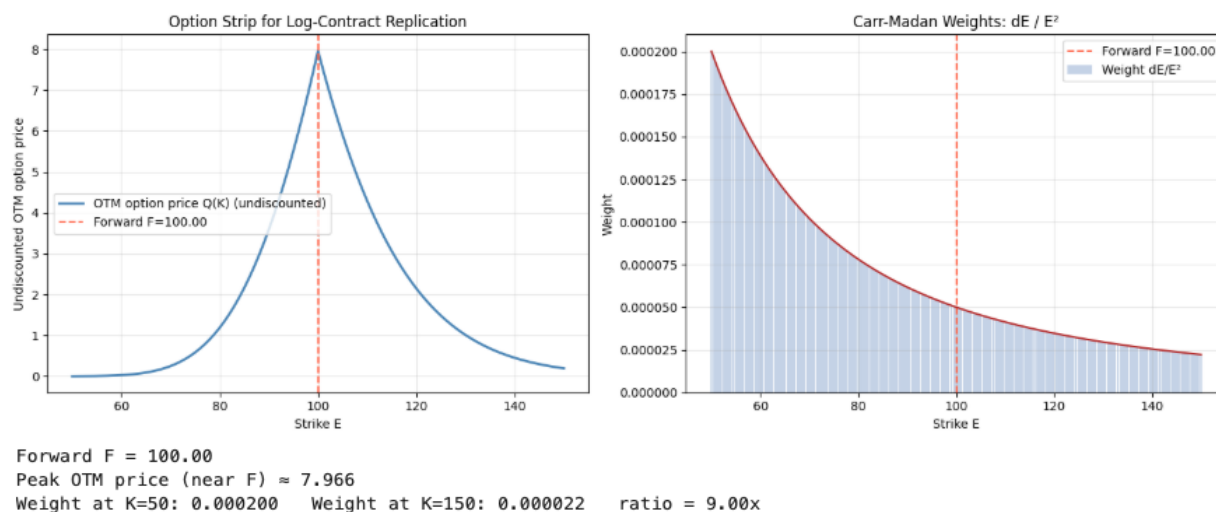
$$VIX^2 \approx \frac{2}{T} \sum_i \frac{\Delta E_i}{E_i^2} Q_i(E_i) - \frac{1}{T} \left(\frac{F}{E_0} - 1\right)^2$$

It is the VIX-style discretization of the continuous option strip.

Common numerical definition: $\Delta E_i = \frac{E_{i+1} - E_{i-1}}{2}$, which is the trapezoidal/midpoint quadrature rule.

3.3 Simulations

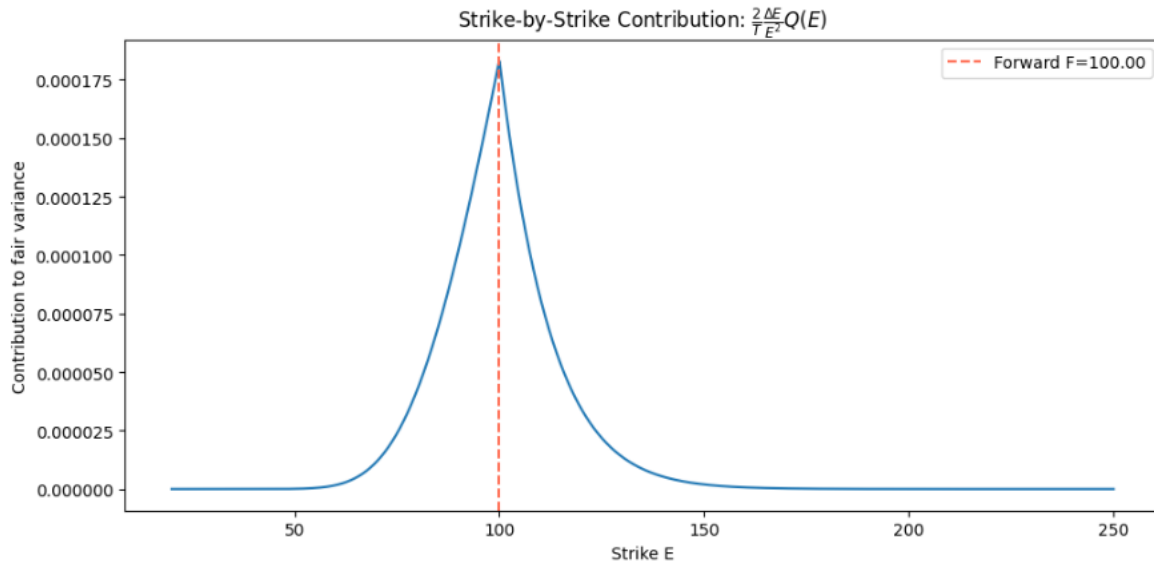
The following plots show the decomposition of the Breeden-Litzenberger / Carr-Madan variance replication strip, the weights, and the combined contribution.



Scenario: $S_0 = 100$, $\mu = 0.0$, $q = 0$, $\sigma = 0.2$, $T = 1.0$

The left panel, the full OTM-option strip, depends on the market, while the right panel, the Carr-Madan weights $\frac{\Delta E_i}{E_i^2}$, is a fixed function of the strike grid. It is the numerical quadrature weight, the discrete approximation to the log-contract curvature kernel. The hump in the left panel occurs near the forward, since near-the-money options carry the most extrinsic value, tapering off as moving toward to OTM. The right panel shows a smooth, monotonically decreasing function of strikes, no peak, no dependence on the forward and no dependence on the smile either. The Carr-Madan weight structurally over-weights low strikes. Actual strike contributions depend on the product $\frac{\Delta E_i}{E_i^2} Q_i(E_i)$.

Under a real negative skew typically observed in equity-index options, this compounds with elevated downside put prices make variance swaps and VIX especially sensitive to the left tail. The discretized bars in the plot are dense, since the simulation here uses a large number of idealized strikes, while in reality, there is a small number of coarse, tradeable strikes.



Scenario: $S_0 = 100$, $\mu = 0.0$, $q = 0$, $\sigma = 0.2$, $T = 1.0$

The entire and actual strike contribution is shown above, combining effects of weights and option values.

4. Future Extensions

- 1) Apply a volatility surface, such as local or stochastic volatility modeling, to simulate dynamics and hedging.
- 2) Use a regime-switching model, such as Markov switching, to simulate the actual volatility switches.
- 3) Introducing a compound Poisson process to evaluate the effect of jumps. The variance swap replication identity is built under the assumption of the continuous diffusion, with no jumps. Jump risk is a source of replication breakdown.
- 4) Considering transaction costs and discrete hedging.